Data Visualisation of Cryptocurrency Market Volatility for Risk Analysis

Dataset: https://www.kaggle.com/datasets/sudalairajkumar/cryptocurrencypricehistory

Student Name: [Your name]

Student ID: [Your ID]

Supervisor Name: [Supervisor's name]

Table of Contents

1. Introduction 3

1.1 Problem Statement 3

1.2 Importance of the Study 3

1.3 Beneficiaries of the Project 3

2. Brief Background 4

2.1 State of the Art 4

2.2 Current Solutions 4

2.3 Limitations of Existing Solutions 5

2.4 Challenges and Difficulties 5

3. Methods 5

3.1 Data Source 5

3.2 Data Preprocessing 6

3.3 Project Approach 6

3.4 Aim and Objectives 6

3.5 Research Questions 7

3.6 Expected Contribution 7

4. Evaluation 8

4.1 Evaluation Method 8

4.2 Comparative Analysis 8

5. Time Plan (Gantt Chart) 9

6. Conclusion 9

References 10

1. Introduction

1.1 Problem Statement

The cryptocurrency market is characterized by the extreme volatility and the absence of visual means of analysing the market risks clearly. Raw data or fixed charts used by the investors can be very misleading as they do not provide the underlying volatility patterns or exposure to risk. Financial visualisations that are traditional are not efficient in communicating changes and offering information about temporal behaviour. This project helps to fill the gap, as it will create a complete visual analytics platform that will reflect the market volatility and risk trends through interactive dashboards. It also tries to interpret complicated numerical information into visual or understandable formats that help investors make quality financial decisions (Chiu et al., 2023).

1.2 Importance of the Study

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```
Found 23 cryptocurrency datasets:

- coin_NEM.csv
- coin_Stellar.csv
- coin_Monero.csv
- coin_Litecoin.csv
- coin_BinanceCoin.csv
- coin_USDCoin.csv
- coin_Dogecoin.csv
- coin_EOS.csv
- coin_XRP.csv
- coin_Iota.csv
- coin_WrappedBitcoin.csv
- coin_Cardano.csv
- coin_ChainLink.csv
- coin_Tether.csv
- coin_Bitcoin.csv
- coin_Uniswap.csv
- coin_CryptocomCoin.csv
- coin_Solana.csv
- coin_Cosmos.csv
- coin_Tron.csv
- coin_Aave.csv
- coin_Polkadot.csv
- coin_Ethereum.csv

Bitcoin Dataset Info:
```

	SNo	Name	Symbol	Date	High	Low	Open	Close	Volume	Marketcap
0	1	Bitcoin	BTC	2013-04-29 23:59:59	147.488007	134.000000	134.444000	144.539993	0.0	1.603769e+09
1	2	Bitcoin	BTC	2013-04-30 23:59:59	146.929993	134.050003	144.000000	139.000000	0.0	1.542813e+09
2	3	Bitcoin	BTC	2013-05-01 23:59:59	139.889999	107.720001	139.000000	116.989998	0.0	1.298955e+09
3	4	Bitcoin	BTC	2013-05-02 23:59:59	125.599998	92.281898	116.379997	105.209999	0.0	1.168517e+09
4	5	Bitcoin	BTC	2013-05-03 23:59:59	108.127998	79.099998	106.250000	97.750000	0.0	1.085995e+09

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2991 entries, 0 to 2990
Data columns (total 10 columns):
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

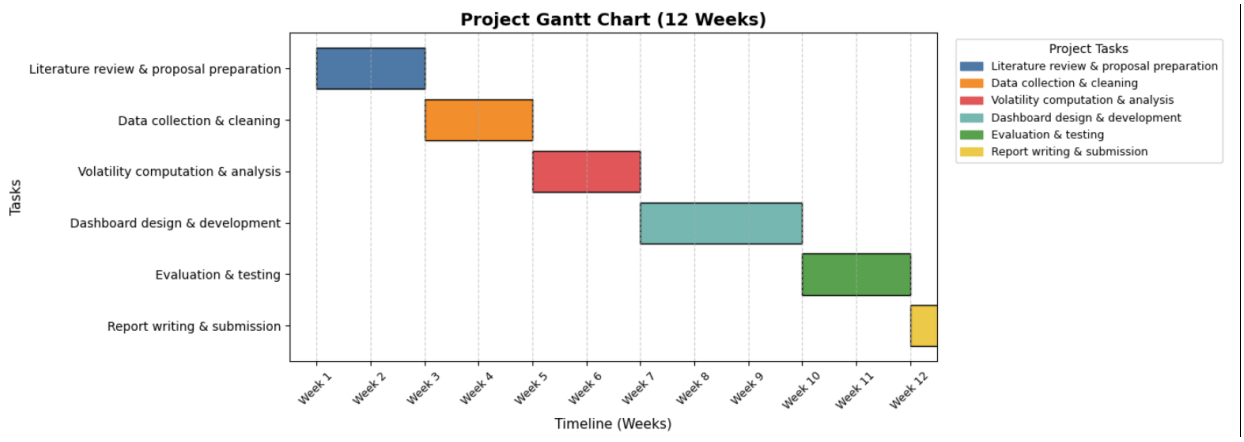
[redacted]

[redacted]

[redacted]

[redacted]

[redacted]



[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]